

Ch 2: Thinking Like an Economist

Models, the circular flow, and the PPF

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

Economists wear two hats

Scientist

Explain how the economy works.
Observe, build a theory, test it
against new data.

Policy adviser

Recommend what to do about it.
Requires facts *and* values.

This chapter: how the science is done, and how to keep the two hats apart.

The scientific method, without a lab

- Same loop as physics or biology: observation, theory, more observation.
- The difference: economists usually cannot run controlled experiments.
- Substitute: **natural experiments**. One state raises its minimum wage; the neighboring state does not. Compare.
- Every science simplifies. A physicist drops a marble and ignores air resistance.
- Economists assume two countries and two goods to study trade. The logic survives when the numbers grow.

Economic model

A simplified representation of part of the economy, built from assumptions, expressed in words, diagrams, or equations.

- A model is judged by whether it answers the question, not by how much detail it keeps.
- The right assumption depends on the question: fixed prices are fine for next month, not for the next decade.

Two models today: the **circular-flow diagram** and the **production possibilities frontier**.

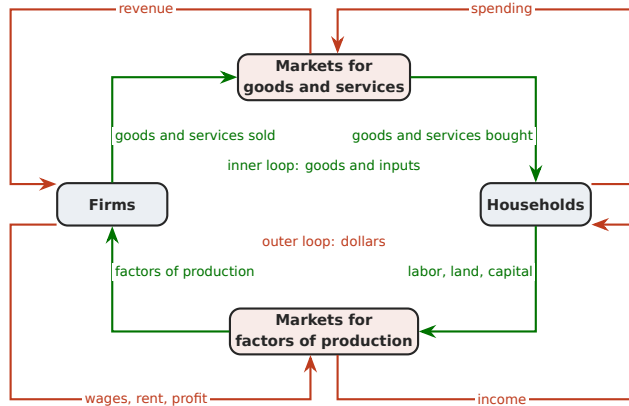
Two decision makers

- **Households:** own labor, land, capital; sell them to firms; buy goods and services.
- **Firms:** hire the factors; produce and sell goods and services.

Two markets

- **Goods and services:** firms sell, households buy.
- **Factors of production:** households sell, firms buy.

The circular-flow diagram



Following a dollar

Question

You pay \$5 for a coffee. Trace the \$5 around the loop. Where does it end up?

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- Spending in the goods market; revenue to the cafe.
- The cafe pays it out as the barista's wage, the landlord's rent, the owner's profit.
- All three are households: the \$5 is now income, and tomorrow it is spent again.

Left out on purpose: government, banks, the rest of the world.

The production possibilities frontier

PPF

The combinations of output an economy can produce with its available factors of production and technology.

Our economy

- Two goods: computers and wheat.
- One resource: labor, 50,000 hours per month.
- 1 computer = 100 hours. 1 ton of wheat = 10 hours.

Question

All 50,000 hours on computers: how many? All on wheat: how many tons?

Step 1: allocate labor, compute output

Point	Labor hours		Output	
	Computers	Wheat	Computers	Wheat (tons)
A	50,000	0		
B	40,000	10,000		
C	25,000	25,000		
D	10,000	40,000		
E	0	50,000		

Step 1: allocate labor, compute output

Point	Labor hours		Output	
	Computers	Wheat	Computers	Wheat (tons)
A	50,000	0	500	0
B	40,000	10,000		
C	25,000	25,000		
D	10,000	40,000		
E	0	50,000		

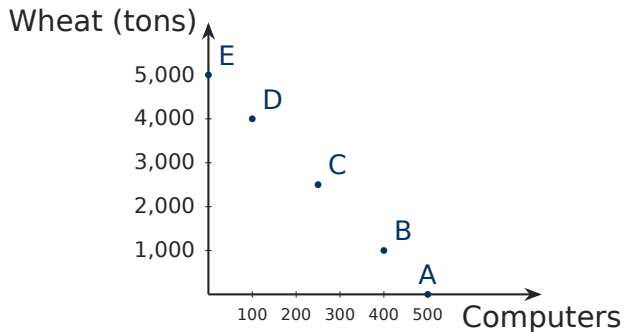
Row A: $50,000/100 = 500$ computers, no wheat.

Step 1: allocate labor, compute output

Point	Labor hours		Output	
	Computers	Wheat	Computers	Wheat (tons)
A	50,000	0	500	0
B	40,000	10,000	400	1,000
C	25,000	25,000	250	2,500
D	10,000	40,000	100	4,000
E	0	50,000	0	5,000

Row A: $50,000/100 = 500$ computers, no wheat. Row C: $25,000/100 = 250$ computers and $25,000/10 = 2,500$ tons.

Step 2: plot the points

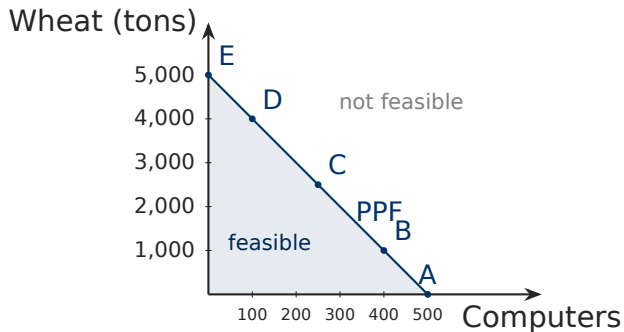


Each point is one way to split the 50,000 hours.

Computers on the horizontal axis, wheat on the vertical.

There are many more feasible splits between the labeled points.

Step 3: the frontier



- **On the line:** efficient. Every hour used; more of one good means less of the other.
- **Inside:** feasible but inefficient. Idle hours.
- **Outside:** not feasible.

Straight line because both goods use labor at fixed rates.

Two more points

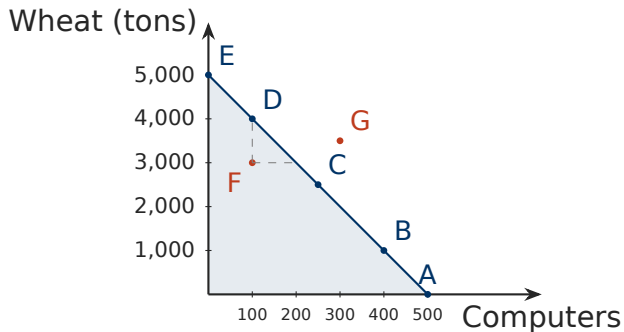
Question

Find each point on the graph. Can the economy produce it? Is it efficient?

- F: 100 computers and 3,000 tons of wheat
- G: 300 computers and 3,500 tons of wheat

Rule: hours needed = $100 \times \text{computers} + 10 \times \text{wheat}$, compared with 50,000.

Two more points: answers



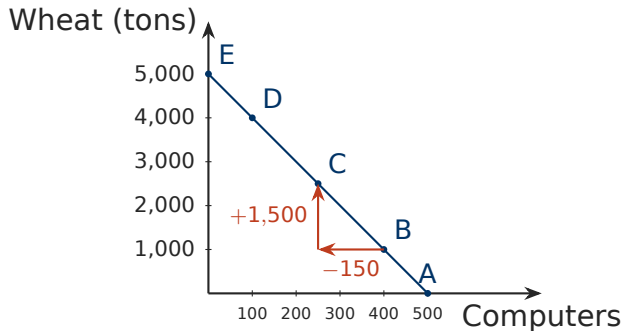
F:

$10,000 + 30,000 = 40,000$ hours. Feasible; 10,000 hours idle. Inside the frontier, inefficient: could move up to D or right to (200, 3,000) for free.

G:

$30,000 + 35,000 = 65,000$ hours. Only 50,000 available. Not feasible.

Opportunity cost is the slope



A to B: -100 computers,
+1,000 tons.

B to C: -150 computers,
+1,500 tons.

Along this PPF

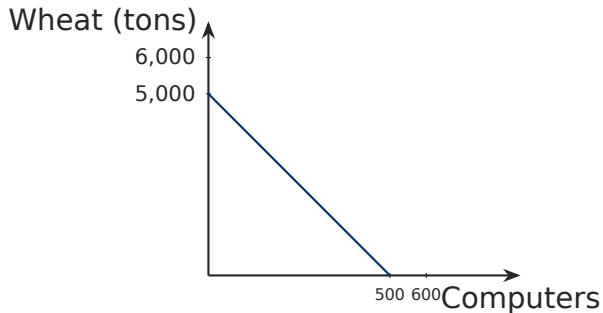
1 computer costs 10 tons
of wheat.

1 ton costs 1/10
computer.

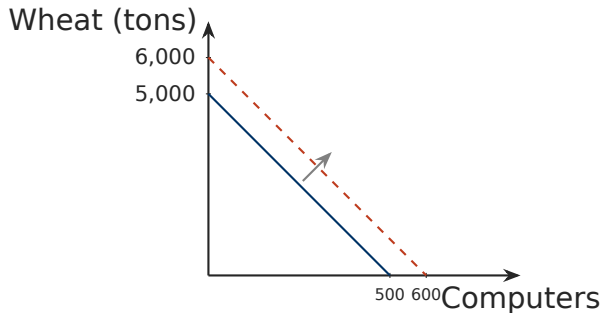
$$|\text{slope}| = 5,000/500 = 10.$$

Growth 1: more labor

Immigration adds 10,000 hours: 60,000 per month.



Growth 1: more labor

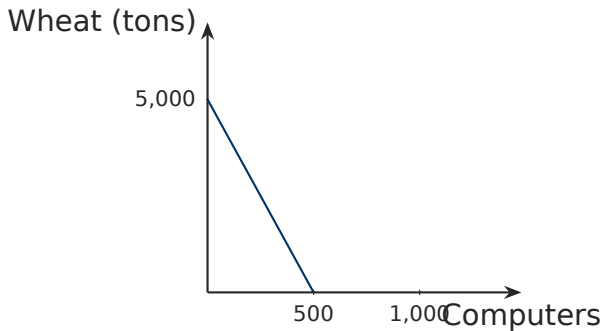


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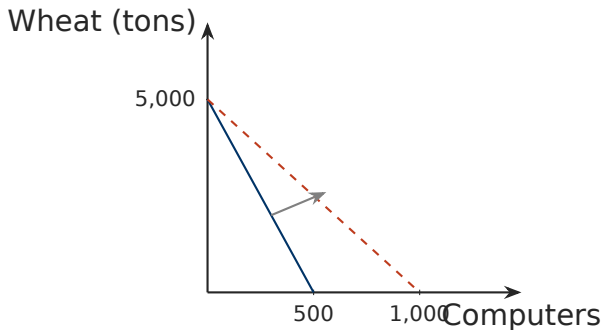
- All computers:
 $60,000/100 = 600$.
- All wheat:
 $60,000/10 = 6,000$.
- Both intercepts up 20 percent: a **parallel** shift.
- Slope unchanged. A computer still costs 10 tons.

Growth 2: better technology

A new process cuts a computer from 100 hours to 50. Wheat unchanged.



Growth 2: better technology



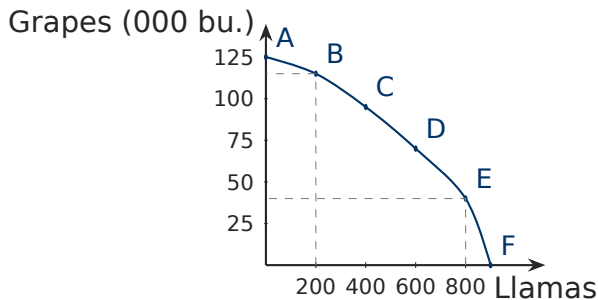
A new process cuts a computer from 100 hours to 50. Wheat unchanged.

- All computers:
 $50,000/50 = 1,000$.
- All wheat: still 5,000.
- The frontier **pivots** on the wheat intercept.
- Slope changes: a computer now costs $50/10 = 5$ tons.

A bowed-out PPF: grapes and llamas

Land in California: part fertile, part rocky. Llamas do fine anywhere; grapes do not.

	Llamas	Grapes (000 bu.)	Cost/llama (bu.)
A	0	125	
B	200	115	50
C	400	95	100
D	600	70	125
E	800	40	150
F	900	0	400



Why the bow?

Question

From A, what do the first 200 llamas cost in grapes? What do the last 100 (E to F) cost?

Why the bow?

Question

From A, what do the first 200 llamas cost in grapes? What do the last 100 (E to F) cost?

- First 200: 10 thousand bushels, 50 per llama. They go on rocky land that grew little.
- Last 100: 40 thousand bushels, 400 per llama. They displace the best vineyard.

Rule

Straight PPF: constant opportunity cost (identical resources).

Bowed-out PPF: increasing opportunity cost (resources differ). The cheap units come first.

Micro and macro

Microeconomics

How households and firms make decisions and how they interact in markets.

Why did rents in Boston rise?

What does a steel tariff do to car prices?

Connected, but different questions and different tools. This course is the micro half.

Macroeconomics

Economy-wide phenomena: inflation, unemployment, growth.

Why does the whole economy grow in some decades and stall in others?

Positive and normative

Positive statement

How the world *is*. Evidence could confirm or refute it.

“A \$20 minimum wage would reduce teen employment.”

Positive does not mean true. “The moon is made of cheese” is positive: testable and false.

Watch for *should*, *ought*, *needs to*.

Normative statement

How the world *ought to be*. Rests on values; evidence alone cannot settle it.

“The minimum wage should be \$20.”

Positive or normative?

Question

- (a) Prices rise when the central bank prints more money.
- (b) The central bank should print less money.
- (c) A tax cut is needed to get the economy moving.
- (d) A higher price of burritos raises the demand for ramen.

Positive or normative?

Question

- (a) Prices rise when the central bank prints more money.
- (b) The central bank should print less money.
- (c) A tax cut is needed to get the economy moving.
- (d) A higher price of burritos raises the demand for ramen.

(a) positive. (b) normative. (c) normative: “needed” is a judgment about goals. (d) positive, whether or not it is true.

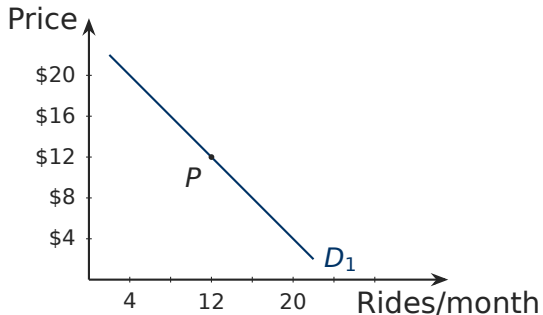
Why economists disagree

1. **Different scientific judgments.** Which theory is right? How big is the effect? Empirical questions, not always settled.
2. **Different values.** Same facts, different weights on winners and losers. No dataset resolves it.
3. **Less disagreement than it looks.** About 93 percent of economists agree that rent ceilings reduce the quantity and quality of housing; 93 percent that tariffs and quotas usually reduce welfare. TV still finds one on each side.

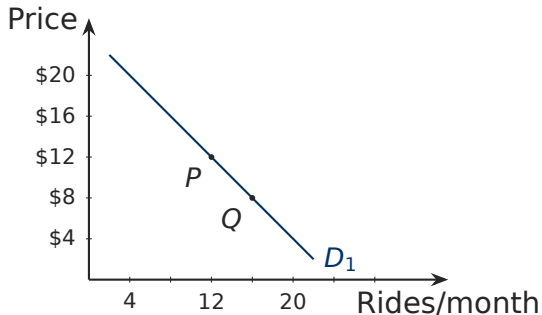
Their advice is one input. The political process makes the call.

Graphing: along the curve versus shifting it

Maya's rides depend on price and income.



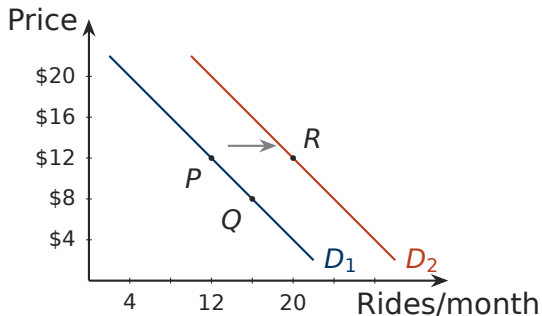
Graphing: along the curve versus shifting it



Maya's rides depend on price and income.

- Price \$12 to \$8, income fixed: **move along** D_1 , P to Q .
Slope = $-4/4 = -1$.

Graphing: along the curve versus shifting it



Maya's rides depend on price and income.

- Price \$12 to \$8, income fixed: **move along** D_1 , P to Q .
Slope = $-4/4 = -1$.
- Income \$2,000 to \$3,000: not on an axis, so the curve **shifts**, D_1 to D_2 ; at \$12, P to R .

"Price fell, so demand rose" is wrong: quantity demanded rose.

Correlation is not causation

Omitted variable

Ice cream sales and drownings rise together every summer. Ice cream does not cause drowning; heat drives both.

Reverse causality

Cities with more police have more crime. Police do not cause crime; crime causes cities to hire police.

Question

Front-row students earn higher grades. A dean assigns everyone a front-row seat. Will grades rise?

Summary

- Models simplify on purpose; test them against data.
- Circular flow: households and firms, two markets, goods one way and dollars the other.
- PPF: on the line efficient, inside inefficient, outside infeasible. Opportunity cost is the slope: 10 tons per computer. More resources shift it out; one-sided technology pivots it.
- Bowed-out PPF: increasing opportunity cost from heterogeneous resources.
- Positive: testable. Normative: value judgment.
- Along a curve (axis variable changes) versus a shift (anything else). Correlation is not causation.